S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026 · CASE STUDY

TRIDENT.

A real-time AI/NLP risk engine with a tactical index rebalancer and a wholesale-banking stress tester

One engine · two strategies · three signals

[Your Full Name]

Indian Institute of Technology Madras

github.com/<your-username>/iitm-<your-name>-hackathon

LIVE SIGNAL · SVB REPLAY, 8 MAR 2023

"SVB Financial announces emergency capital raise after $1.8 billion loss on bond portfolio sale"

Entity	**SIVB · SVB Financial · alias 0.94**
Event	**Credit Event · liquidity**
Sentiment	**−0.62**
Impact	**9 / 10**
Gate	**TRIGGERED · 2 sources**
Engine	**1.43 ms p50**

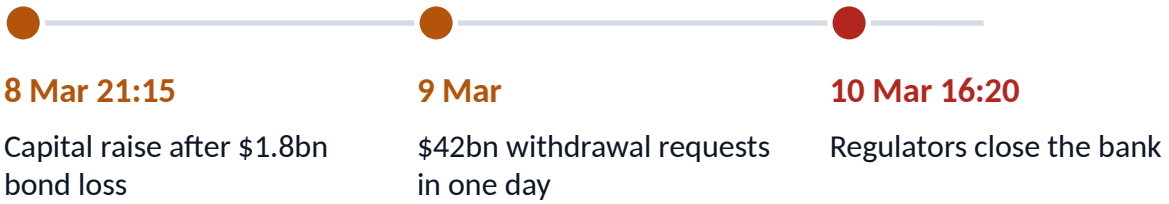
PROBLEM & APPROACH

Risk arrives as text hours before it arrives as data

The problem

- Market-moving news, filings and posts are unstructured; risk systems read structured feeds.
- Free sources lag (GDELT 15 min, NewsAPI free 24 h) and are full of duplicates, rumors and ambiguous names.
- A useful signal must be fast, calibrated, explainable and tied to a decision.

SVB, March 2023 (UTC)



Our approach

Resolve

Map every item to a company, a sector or the market; reject ambiguous names with a logged reason.

Score

Target-aware sentiment, event class with abstain, impact calibrated to realized moves; confidence and evidence on each.

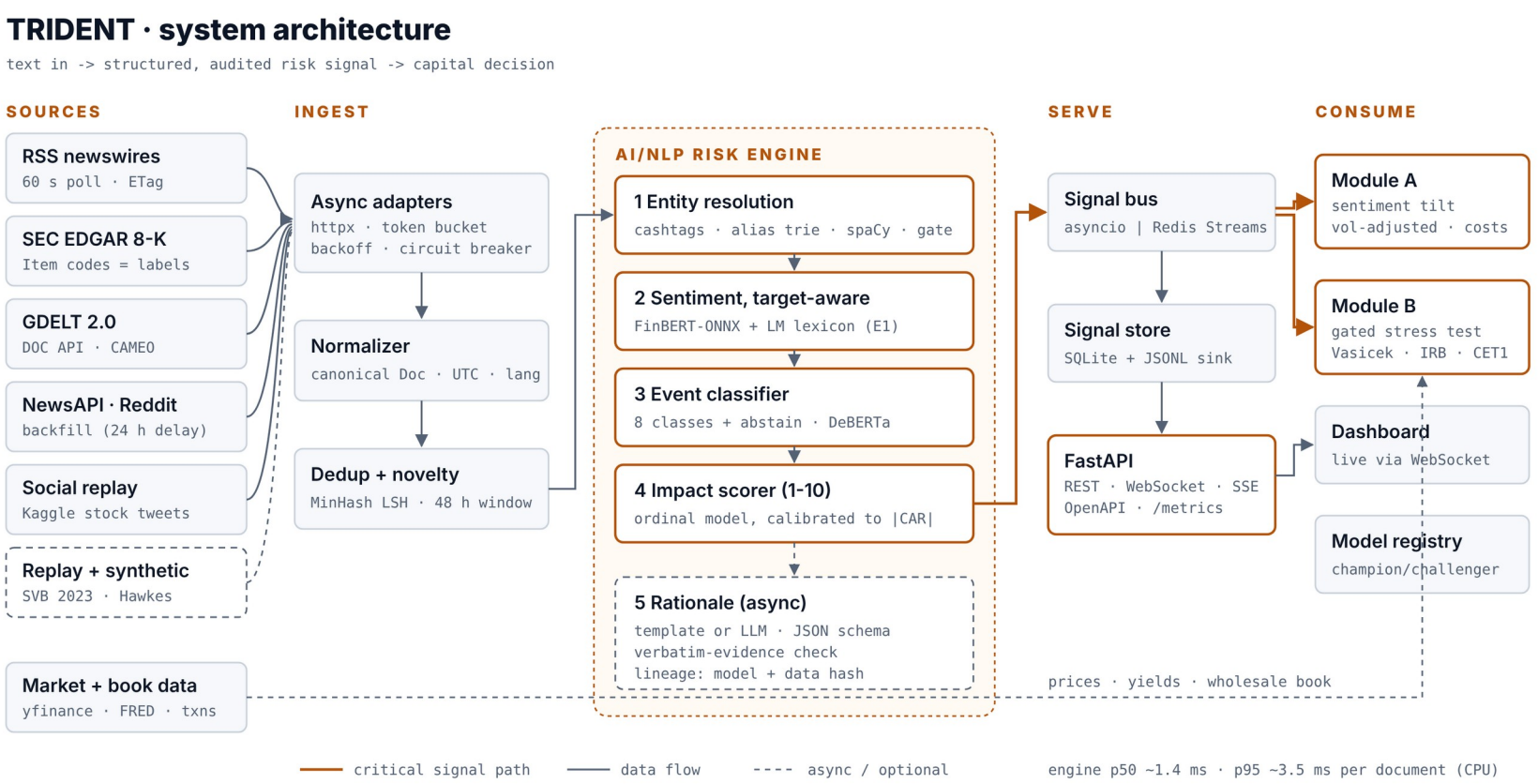
Act

Module A tilts a 15-stock index (minutes to days). Module B stress-tests a wholesale book and reports CET1 (capital).

Both modules built: A consumes sentiment, B consumes event + impact, so all three fields are exercised.

SYSTEM DESIGN

Simple to run on a laptop, with a clear path to bank scale



Replay or live

Deterministic replay of SVB-2023 and COVID-2020 means the demo needs no keys and no network; live mode polls RSS, SEC EDGAR, GDELT, NewsAPI and Reddit.

Pluggable bus

In-process asyncio queue by default; Redis Streams with one setting; Kafka partitioned by ticker behind the same interface.

Lineage on every signal

Model versions, data hash and replay id travel with each signal; a champion/challenger registry logs every model change.

Measured: engine 1.43 ms p50 / 3.29 ms p95 per document · Module A rebalance 0.34 ms · Module B full revaluation 2.75 ms · 855 docs/s on one CPU process

IMPLEMENTATION HIGHLIGHTS

Four design choices that separate signal from noise

Target-aware sentiment

Finance lexicon with Loughran-McDonald words, negation and delta phrases, blended with FinBERT exported to ONNX INT8. Scored per entity: “Microsoft beats Google” is + for MSFT, − for GOOGL.

VADER scores “Profit plunges after default” as +0.44. TRIDENT: −0.91.

Events with an abstain class

8 classes plus Other, weak labels from keywords, 8-K Item codes and GDELT CAMEO, distilled into DeBERTa. A champion/challenger registry only promotes a model that beats the incumbent.

Synthetic-only challengers were rejected (F1 0.48 vs 0.65).

Impact calibrated to realized moves

An ordinal model maps severity, |sentiment|, novelty, burst velocity, source reliability and magnitude cues to deciles of |CAR[0,+1]| from a market-model event study.

Out-of-sample Spearman 0.20 (5-seed mean 0.23).

Module B, the way a bank measures it

Vasicek PD migration, duration-convexity, Greeks and Black-Scholes, Basel IRB RWA and CET1, stressed-correlation VaR, reverse stress test. A two-source gate stops single headlines.

Credit I=9: CET1 13.0% → 8.4%.

Python 3.11 · FastAPI · Pydantic · scikit-learn · SciPy · cvxpy · transformers + onnxruntime · spaCy · ECharts · 55 tests incl. Hypothesis property tests

KEY RESULTS

Measured against baselines, with the data source named

1.43 ms

Engine latency p50

p95 3.29 ms · 855 docs/s

0.65 vs 0.47

Sentiment F1 vs VADER

authored 180-headline set (optimistic)

8.4%

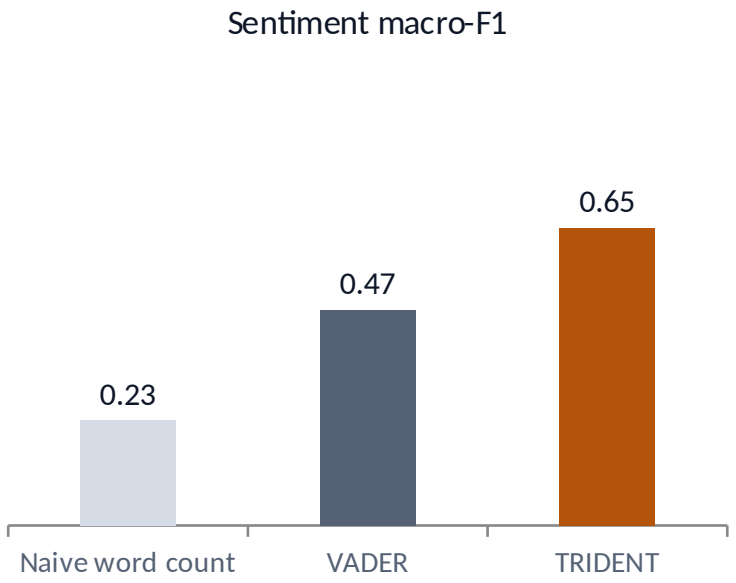
CET1 after Credit I=9

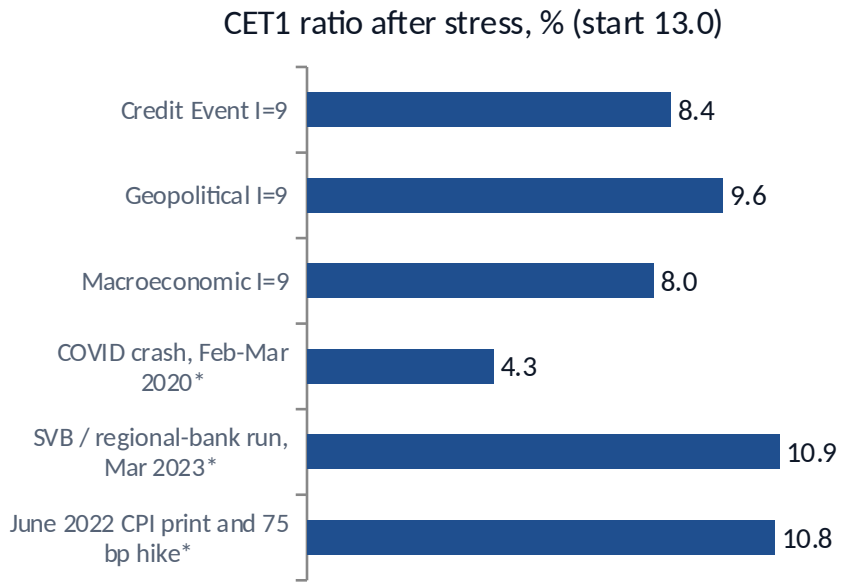
from 13.0% · P&L -80m

58/58

Behavioral tests passed

entity precision 1.00 · recall 0.98





Module A, out of sample

0.95

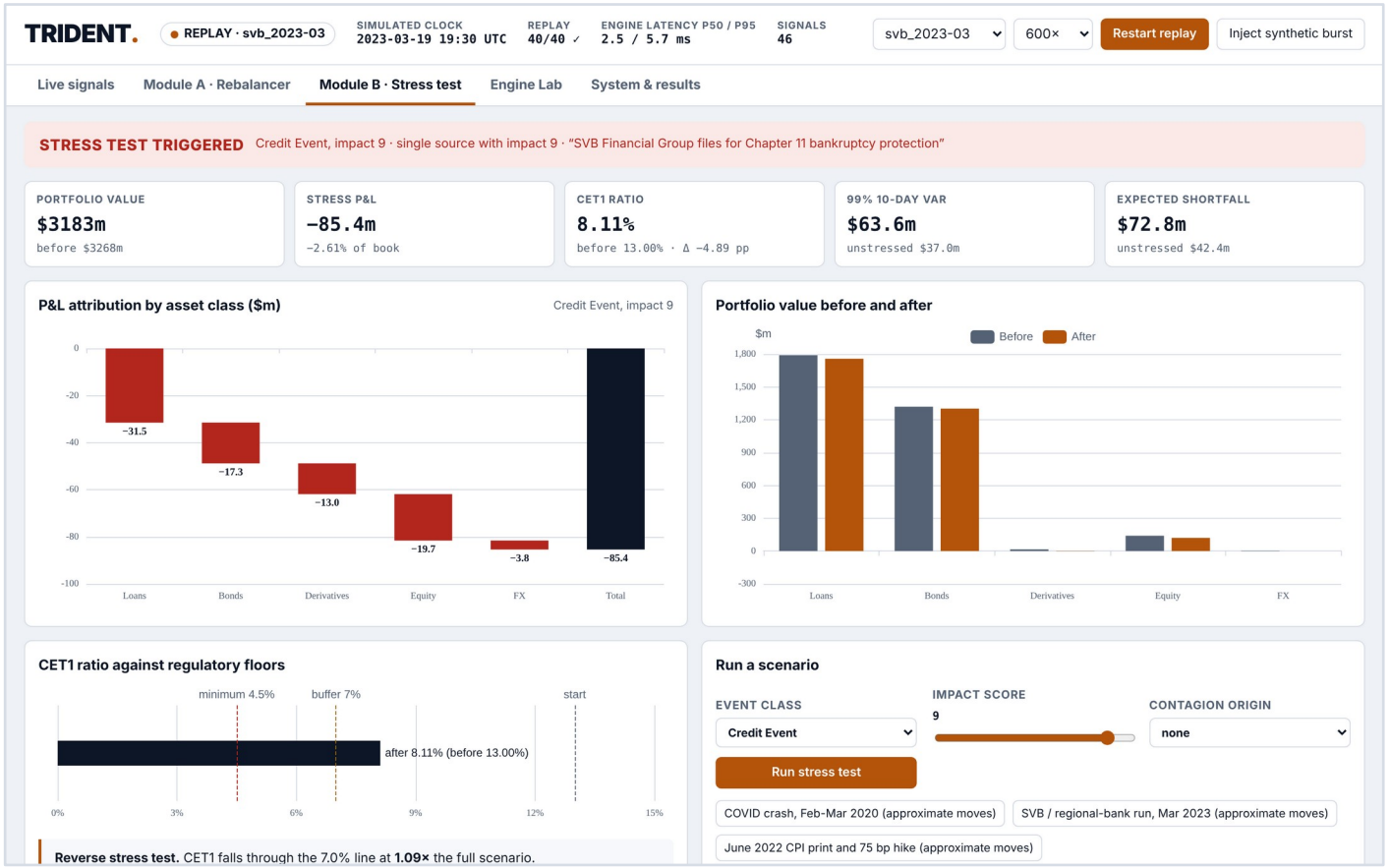
net information ratio (gross 1.23), 2024-2025, parameters chosen on earlier years

Across 5 seeds: net IR 0.29 ± 0.62, IC 0.064. The signal is real; costs decide whether it pays.

Data: sentiment and entities on authored 180-headline set (optimistic); impact and Module A on synthetic known-answer market; Module B on the synthetic wholesale book. Regenerate on real data with scripts/run_all_results.py. *approximate historical moves.

DOMAIN IMPACT

From headline to capital decision in one replay



SVB replay, 8 March 2023 (UTC)

21:15 First headline, impact 8, one source — HELD

21:40 Second wire confirms the story — TRIGGERED

→ P&L -90m · CET1 13.0% → 7.9% — CAPITAL VIEW

Credit surveillance

Obligors flagged within minutes, ranked by calibrated impact, evidence attached.

Portfolio managers

A cost-aware tilt with caps and bands, not headline chasing.

Risk & treasury

Event-triggered CET1, VaR and reverse stress for the risk committee.

What we know is weak, and what we would build next

Limitations

- Free sources lag by minutes to a day; engine speed is not end-to-end speed.
- Sentiment and entity scores use an authored set; real PhraseBank and labeled headlines are the numbers to report.
- Template synthetic text is easy to detect (AUC 0.95); it is capped at 30% of training and never used for evaluation.
- Impact and Module A were validated on a known-answer market; real-price results depend on data we fetch locally.
- IRB without maturity adjustment; Greeks are linear except the options book.

Next steps

- Fine-tune FinBERT and DeBERTa on real labels; serve ONNX INT8.
- Indian market: NSE universe, Indian regulatory disclosures, Hindi and regional-language news via GDELT.
- LLM-generated synthetic data that passes adversarial validation.
- Sector contagion through a learned correlation network; online recalibration of impact.
- Rating-surveillance view: map signals to obligors in a credit portfolio.

TRIDENT: text in, capital decision out, every step auditable.